The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF).

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Five-Step Abstract Ontology Replacement Merger Protocol

We define the formal sequencing bounds for executing single-pass ontology alignment across distinct agent nodes using pre-resident grounding exemplars.

3.1 The Local Grounding Initialization Constraint

Let $\mathcal{E}_k$ represent the local extensional ground plane stored inside an agent's `exs.pl` file framework. The initialization constraint mandates that all historical complex exemplars must be fully compiled and serialized prior to data transport:

$$\mathcal{E}_{\text{target}} \equiv \langle C_1', C_2', \ldots, C_n' \rangle \in \text{Local Memory} \tag{2}$$

This baseline provides the immutable structural stencils required to interrogate incoming models.

3.2 The Single-Pass Interrogation and Realignment Sequence

Upon receiving the source agent's abstract tree model via the Model Context Protocol (MCP), the target node executes a 5-step replacement sequence:

1. **Logic Basis Consensus:** The target verifies the incoming `kb.pl` rules against master invariants to clear a shared logical foundation.

2. **Model Payload Buffering:** The target agent copies the source abstract decision tree $\mathcal{T}_{\text{source}}$ directly into a protected memory cache.

3. **Blind Intensional Interrogation:** The target unrolls its local complex exemplars, strips out the explicit classification type names to form blind query vectors $\boldsymbol{v}_{\text{query}}$, and evaluates them across both models. Let $\mathcal{H}_T$ represent the target tree traversal function and $\mathcal{H}_S$ represent the source copy function:

$$\mathcal{H}_T(\boldsymbol{v}_{\text{query}}) \implies \mathcal{L}_{\text{target}}, \quad \mathcal{H}_S(\boldsymbol{v}_{\text{query}}) \implies \mathcal{L}_{\text{source}} \tag{3}$$

   If a structural nomenclature variance occurs ($\mathcal{L}_{\text{target}} \neq \mathcal{L}_{\text{source}}$), the target captures the discrepancy once, maps the corrected source name to the feature set, and locks the repaired exemplar into the local `exs.pl` file matrix.

4. **Inductive Retraining Over Delta Buffer:** The target executes an immediate local inductive training pass over the updated, repaired exemplar list, absorbing the merged terms.

5. **Ontology Substitution Merger:** The newly retrained abstract model replaces the old target tree configuration, achieving complete semantic realignment across the multi-agent perimeter.